Verona through Rome

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1 Introduction

We present *Roma*^{M}, a more complete, fully formalized version of *Rome*^{M} [Nguyen 2026].

2 Model

The configuration remains unchanged from the report. We make one change to the semantics, specifically within `makeRegion`:

MAKE-REGION

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \langle (S :: F), H\rangle.freshObjectId = oid' \qquad \langle (\mathbf{S} :: \mathbf{F}), \mathbf{H}\rangle.\mathbf{freshRegionId} = \mathbf{rid'} \\ R' = (oid', [oid' \mapsto \emptyset], Closed) \qquad F' = (rid, bvar, omap, vmap[\mathtt{x} \mapsto \mathsf{RId}(rid')], fid) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeRegion x}} \langle S :: F', H[rid \mapsto R']\rangle}$$

In the report, the `regionId` would coincide with the `objectId` of the bridge object, but here we've chosen to completely decouple them. The original reasoning behind this was to future-proof for deallocation, but in writing this we found out that the argument didn't hold, and it would've been fine to leave `makeRegion` as before. Luckily, this doesn't change much, it only means that we need to prove regions are unique separately from objects, as opposed to using a combined argument before.

3 Invariants

Definition 3.1 (L1). No two objects in the configuration share the same identifier.

$$\langle S, H\rangle \vdash L1 \triangleq \langle S, H\rangle.objectIds \text{ is duplicate-free.}$$

Definition 3.2 (L2). Every frame on the stack refers to an open region.

$$\langle S, H\rangle \vdash L2 \triangleq \forall frame \in S : \exists region : [H(frame.regionId) = region \wedge region.status = Open]$$

Definition 3.3 (H1). A region always contains its own bridge object.

$$\langle S, H\rangle \vdash H1 \triangleq \forall region \in H.regions : [region.bridgeObjectId \in dom(region.objMap)]$$

Definition 3.4 (H2). There is at most one reference to a region from anywhere in the configuration.

$$\langle S, H\rangle \vdash H2 \triangleq \forall rid : [S.refs.count(\mathsf{RId}(rid)) + H.refs.count(\mathsf{RId}(rid)) \leq 1]$$

Definition 3.5 (H3). A reference stored inside a region resolves back into that same region.

$$\begin{aligned}\langle S, H\rangle \vdash H3 \triangleq \forall rid, oid, region : [&H(rid) = region \wedge \mathsf{OId}(oid) \in region.refs \implies \\ &\mathsf{OId}(oid).loc(\langle S, H\rangle) = \mathsf{Rgn}(rid)]\end{aligned}$$

Definition 3.6 (S1). No two frames on the stack share a region.

$$\langle S, H\rangle \vdash S1 \triangleq \forall f_1, f_2 \in S : [f_1.regionId = f_2.regionId \implies f_1 = f_2]$$

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Definition 3.7 (S2). A reference from a frame to a temporary object resolves to that frame or an earlier one.

$$\langle S,H\rangle \vdash S2 \triangleq \forall frame \in S, \mathsf{OId}(oid) \in frame.refs, fid' : [\mathsf{OId}(oid).loc(\langle S,H\rangle) = \mathsf{Stk}(fid') \implies fid' \leq frame.fid]$$

Definition 3.8 (S3). A reference from a frame to a region object resolves to that frame's region or the region of an earlier frame.

$$\langle S,H\rangle \vdash S3 \triangleq \forall frame \in S, \mathsf{OId}(oid) \in frame.refs, rid' : [\mathsf{OId}(oid).loc(\langle S,H\rangle) = \mathsf{Rgn}(rid') \implies \exists frame' \in S : [frame'.regionId = rid' \land frame'.fid \leq frame.fid]]$$

We add two new invariants:

Definition 3.9 (HS1). Every reference to an object must be to an object that exists in the configuration.

$$\langle S,H\rangle \vdash HS1 \triangleq \forall oid : [\mathsf{OId}(oid) \in \langle S,H\rangle.refs \implies oid \in \langle S,H\rangle.objectIds]$$

Definition 3.10 (HS2). Every reference to a region must be to a region that exists in the configuration.

$$\langle S,H\rangle \vdash HS2 \triangleq \forall rid : [\mathsf{RId}(rid) \in \langle S,H\rangle.refs \implies rid \in dom(H)]$$

These came up in the process of proving *S2* and *S3* for `makeObjStack`, `makeObjRegion`, and `makeRegion`. This stems from the fact that our `freshObjectId` and `freshRegionId` functions only take into account the `Ids` used as keys when creating a new `Id`. Thus, without this invariant, it is technically possible[1] for a configuration to contain a dangling reference to an `objectId` or `regionId` that does not exist within the configuration, but is then inserted via one of the three `make` operations.

Definition 3.11 (ValidConfig).

$$\textsc{Valid}\quad \frac{\langle S,H\rangle \vdash H3 \quad \begin{array}{c}\langle S,H\rangle \vdash L1\\ \langle S,H\rangle \vdash S1\end{array} \quad \begin{array}{c}\langle S,H\rangle \vdash L2\\ \langle S,H\rangle \vdash S2\end{array} \quad \begin{array}{c}\langle S,H\rangle \vdash H1\\ \langle S,H\rangle \vdash S3\end{array} \quad \begin{array}{c}\langle S,H\rangle \vdash H2\\ \langle S,H\rangle \vdash HS1\end{array} \quad \langle S,H\rangle \vdash HS2}{\langle S,H\rangle \vdash ValidConfig}$$

`RuntimeConfig.start` (one root region, one root frame) satisfies *ValidConfig*, and every one of the nine mutation operations of §B preserves it; consequently every reachable configuration of the operational semantics is a *ValidConfig*.

4 Reachability

We introduce the notion of reachability. It is the same term as in the original report, but the definition here covers a wider spectrum.

Definition 4.1 (Stack-reachable). An object or region is stack-reachable if it is frame-reachable from some frame in the stack.

$$\frac{\langle S,H\rangle \vdash FReachable(\_, a)}{\langle S,H\rangle \vdash SReachable(a)}$$

[1] Although it isn't possible within the context of an operational semantics starting with the starting configuration, which is why we were able to prove it without altering the operational semantics.

Definition 4.2 (Frame-reachable). An object or region is frame-reachable if there is a path from a frame to said object or region.

$$\textsc{var}\quad \frac{F = (\_, \_, \_, vmap, fid) \qquad vmap(\_) = a}{\langle S :: F, H\rangle \vdash FReachable(fid, a)} \qquad \textsc{bridge}\quad \frac{F = (rid, \_, \_, \_, fid) \qquad H(rid) = (boid, \_, \_)}{\langle S :: F, H\rangle \vdash FReachable(fid, \mathsf{OId}(boid))}$$

$$\textsc{step}\quad \frac{\langle S, H\rangle \vdash FReachable(fid, a) \qquad \langle S, H\rangle \vdash a \rightarrow b}{\langle S, H\rangle \vdash FReachable(fid, b)}$$

Definition 4.3 (Region-reachable). An object or region is region-reachable if there is a path from the bridge object to said object or region.

$$\textsc{bridge}\quad \frac{H(rid) = (boid, \_, \_)}{\langle S, H\rangle \vdash RReachable(rid, \mathsf{OId}(boid))} \qquad \textsc{step}\quad \frac{\langle S, H\rangle \vdash RReachable(rid, a) \qquad \langle S, H\rangle \vdash a \rightarrow b}{\langle S, H\rangle \vdash RReachable(rid, b)}$$

Definition 4.4 (Reachable-step). A one-step reachable relation between objects and regions. An object or region is one-step reachable from another object if that object contains a reference to said object or region. An object is one-step reachable from a region if the region is closed and the object is the region's bridge object.

$$\textsc{obj-step}\quad \frac{\mathsf{OId}(oid).objAt(\langle S, H\rangle) = obj \qquad b \in obj.refs}{\langle S, H\rangle \vdash \mathsf{OId}(oid) \rightarrow b} \qquad \textsc{region-step}\quad \frac{H(rid) = (boid, \_, Closed)}{\langle S, H\rangle \vdash \mathsf{RId}(rid) \rightarrow \mathsf{OId}(boid)}$$

Theorem 4.5. *Region-reachability is the transitive closure of the reachable-step from the region's bridge object.*

$$\langle S, H\rangle \vdash RReachable(rid, a) \iff \exists boid : [H(rid) = (boid, \_, \_) \land \mathsf{OId}(boid) \xrightarrow{*} a]$$

Theorem 4.6. *Frame-reachability is the transitive closure of the reachable-step from the frame.*

$$\langle S, H\rangle \vdash FReachable(fid, a) \iff \exists start : [\langle S, H\rangle \vdash FRoot(fid, start) \land start \xrightarrow{*} a]$$

Definition 4.7 (Frame-root). The root coming from a frame can be a reference stored by a variable or the bridge object of that frame's region.

$$\textsc{var}\quad \frac{F = (\_, \_, \_, vmap, fid) \qquad vmap(\_) = a}{\langle S :: F, H\rangle \vdash FRoot(fid, a)} \qquad \textsc{bridge}\quad \frac{F = (rid, \_, \_, \_, fid) \qquad H(rid) = (boid, \_, \_)}{\langle S :: F, H\rangle \vdash FRoot(fid, \mathsf{OId}(boid))}$$

4.1 Suspended regions

Definition 4.8 (FrameReachableAtLaterFrame). A configuration satisfies *FrameReachableAtLaterFrame* if, whenever a frame F is earlier on the stack than a frame F', any object located in F's region that is frame-reachable from F' is already frame-reachable from F itself.

$$\begin{aligned} FRALF(\langle S, H\rangle) \iff\ & \forall F, F' \in S.\ F.fid < F'.fid \implies \\ & \forall oid.\ \mathsf{OId}(oid).loc(\langle S, H\rangle) = \mathsf{Rgn}(F.regionId) \implies \\ & \langle S, H\rangle \vdash FReachable(F'.fid, \mathsf{OId}(oid)) \implies \langle S, H\rangle \vdash FReachable(F.fid, \mathsf{OId}(oid)) \end{aligned}$$

Theorem 4.9 (FRALF is preserved). *Let cmd be any of the nine operations of §B. If $\langle S, H\rangle$ is a ValidConfig satisfying FRALF, and $\langle S, H\rangle \xrightarrow{cmd} \langle S', H'\rangle$, then $\langle S', H'\rangle$ also satisfies FRALF.*

Theorem 4.10 (Suspended-region stack-reachability is invariant). *Let cmd be any of the nine operations of §B, and let $\langle S, H\rangle$ be a ValidConfig satisfying FRALF with $\langle S, H\rangle \xrightarrow{cmd} \langle S', H'\rangle$. For any suspended frame $F \in S$ and any object oid located in F's region,*

$$\langle S, H\rangle \vdash SReachable(\mathsf{OId}(oid)) \iff \langle S', H'\rangle \vdash SReachable(\mathsf{OId}(oid))$$

5 Related Work

6 Conclusion

A Configuration

$$\begin{aligned}
RegionId, ObjectId, Index &\subseteq Int\\
VarName, FieldName &\subseteq String\\[1ex]
RuntimeConfiguration &\triangleq Heap \times Stack\\[1ex]
Heap &\triangleq RegionId \rightharpoonup Region\\
Region &\triangleq ObjectId \times ObjMap \times Status\\
Status &\triangleq Open \mid Closed\\[1ex]
Stack &\triangleq List\ Frame\\
Frame &\triangleq RegionId \times VarName \times ObjMap \times VarMap \times Index\\[1ex]
ObjMap &\triangleq ObjectId \rightharpoonup FieldMap\\[1ex]
Object &\triangleq FieldMap\\
FieldMap &\triangleq FieldName \rightharpoonup Reference\\
VarMap &\triangleq VarName \rightharpoonup Reference\\[1ex]
Reference &\triangleq \mathsf{RId}(RegionId) \mid \mathsf{OId}(ObjectId)
\end{aligned}$$

B Operational Semantics

VAR-ASGN-STACK

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \epsilon \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x := y.f}} \langle S :: (rid, bvar, omap, vmap[\mathsf{x} \mapsto \mathsf{OID}(oid)], fid), H\rangle}$$

VAR-ASGN-BRIDGE

$$\frac{\begin{array}{c} F = (rid, \mathsf{x}, \_, \_, \_) \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \\ \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(rid) = (boid, omap, status) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x := y.f}} \langle (S :: F), H[rid \mapsto (oid, omap, status)]\rangle}$$

FIELD-ASGN-STACK

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \qquad \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Stk}(fid) \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid') \qquad omap(oid) = obj \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: (rid, bvar, omap[oid \mapsto obj[\mathsf{f} \mapsto OId(oid')]], vmap, fid)), H\rangle}$$

FIELD-ASGN-REGION

$$\frac{\begin{array}{c} F = (rid, \_, \_, \_, \_) \qquad resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \\ \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid') \\ \mathsf{OId}(oid').loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(rid) = (boid, omap, Open) \qquad omap(oid) = obj \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: F), H[rid \mapsto (boid, omap[oid \mapsto obj[\mathsf{f} \mapsto \mathsf{OId}(oid')]], Open)]\rangle}$$

SWAP-STACK

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{x}ref \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \\ resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{y}fref \qquad \mathsf{OId}(\mathsf{y}oid).loc(\langle (S :: F), H\rangle) = \mathsf{Stk}(fid) \\ omap(\mathsf{y}oid) = obj \qquad omap' = omap[\mathsf{y}oid \mapsto obj[\mathsf{f} \mapsto \mathsf{x}ref]] \\ vmap' = vmap[\mathsf{x} \mapsto \mathsf{y}fref] \qquad F' = (rid, bvar, omap', vmap') \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: F'), H\rangle}$$

SWAP-REGION-OBJECT

$$\frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{OId}(\mathsf{x}oid) \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \\ resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{y}fref \qquad \{\mathsf{OId}(\mathsf{x}oid), \mathsf{OId}(\mathsf{y}oid)\}.loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \qquad obj' = obj[\mathsf{f} \mapsto \mathsf{OId}(\mathsf{x}oid)] \\ R' = (boid, romap[\mathsf{y}oid \mapsto obj'], Open) \qquad F' = (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{y}fref]) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: F'), H[rid \mapsto R']\rangle}$$

SWAP-REGION-REGION

$$\frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{RId}(\mathsf{x}rid) \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{y}fref \\ \mathsf{OId}(\mathsf{y}oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(xrid) = (\_, \_, Closed) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \qquad obj' = obj[\mathsf{f} \mapsto \mathsf{RId}(\mathsf{x}rid)] \\ R' = (boid, romap[\mathsf{y}oid \mapsto obj'], Open) \qquad F' = (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{y}fref]) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: F'), H[rid \mapsto R']\rangle}$$

SWAP-REGION-BRIDGE

$$\frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad \mathsf{x} = bvar \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{yf}oid) \\ \{\mathsf{OId}(\mathsf{y}oid), \mathsf{OId}(\mathsf{yf}oid)\}.loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \\ obj' = obj[\mathsf{f} \mapsto \mathsf{OId}(boid)] \qquad R' = (\mathsf{yf}oid, romap[\mathsf{y}oid \mapsto obj'], Open) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x.f\ :=\ y}} \langle (S :: F), H[rid \mapsto R']\rangle}$$

ENTER

$$\frac{\begin{array}{c}F = (rid, bvar, fomap, vmap, fid) \qquad resolveFA(\mathsf{x.f}, \langle S :: F, H\rangle) = \mathsf{RId}(rid') \\ H(rid') = (boid, romap, Closed) \qquad F' = (rid', \mathsf{a}, \emptyset, \emptyset, fid + 1)\end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{enter x.f a}} \langle (S :: F :: F'), H[rid' \mapsto (boid, romap, Open)]\rangle}$$

EXIT

$$\frac{F = (rid, \_, \_, \_, \_) \qquad H(rid) = (boid, omap, Open)}{\langle (S :: F' :: F), H\rangle \xrightarrow{\texttt{exit}} \langle S :: F', H[rid \mapsto (boid, omap, Closed)]\rangle}$$

MAKE-OBJ-STACK

$$\frac{F = (rid, bvar, omap, vmap, fid) \qquad \langle (S :: F), H\rangle.freshObjectId = oid}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeObjStack x}} \langle S :: (rid, bvar, omap[oid \mapsto \emptyset], vmap[\mathsf{x} \mapsto \mathsf{OId}(oid)], fid), H\rangle}$$

MAKE-OBJ-REGION

$$\frac{\begin{array}{c}F = (rid, bvar, fomap, vmap, fid) \qquad \langle (S :: F), H\rangle.freshObjectId = oid \\ H(rid) = (boid, romap, Open) \qquad R = (boid, romap[oid \mapsto \emptyset], Open)\end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeObjRegion x}} \langle S :: (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{OId}(oid)], fid), H[rid \mapsto R]\rangle}$$

MAKE-REGION

$$\frac{\begin{array}{c}F = (rid, bvar, omap, vmap, fid) \\ \langle (S :: F), H\rangle.freshObjectId = oid' \qquad \langle (S :: F), H\rangle.freshRegionId = rid' \\ R' = (oid', [oid' \mapsto \emptyset], Closed) \qquad F' = (rid, bvar, omap, vmap[\mathsf{x} \mapsto \mathsf{RId}(rid')], fid)\end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeRegion x}} \langle S :: F', H[rid \mapsto R']\rangle}$$

MERGE

$$\frac{\begin{array}{c}F = (rid, bvar, omap, vmap, fid) \\ vmap(\mathsf{x}) = \mathsf{RId}(rid') \qquad H(rid) = (boid, romap, Open) \qquad H(rid') = (boid', romap', Closed) \\ F' = (rid, bvar, omap, vmap[\mathsf{x} \mapsto boid'], fid) \qquad R = (boid, romap \cup romap', Open)\end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{merge x}} \langle S :: F', H[rid' \mapsto \epsilon][rid \mapsto R]\rangle}$$

Definition B.1 (Name). A configuration σ satisfies P if

$$\forall x. \quad \big|\{\, y \mid y \in X(\sigma),\ y = x \,\}\big| \le 1.$$

Definition B.2 (Name).

$$\forall x \in X(\sigma).\ \exists y.\ P(x, y) \land Q(y).$$

Definition B.3 (Step relation).

$$\frac{P(\sigma, a) \qquad Q(a, f, b)}{\sigma \vdash a \xrightarrow{f} b}$$

Definition B.4 (Reachable).

$$Reachable(\sigma, i, b) \iff \exists a.\ Root(\sigma, i, a) \land \sigma \vdash a \xrightarrow{*} b.$$

Definition B.5 (Two-branch step relation).

STEP-OBJECT

$$\frac{P(\sigma, a) = c \qquad c.f = b}{\sigma \vdash a \xrightarrow{r} b}$$

STEP-REGION

$$\frac{Q(\sigma, a) = c \qquad guard(c) \qquad b = target(c)}{\sigma \vdash a \xrightarrow{r} b}$$

Definition B.6 (Inductively-closed reachability).

$$\frac{x \in X(\sigma) \qquad P(x) = a}{Reach(\sigma, i, a)}\ \text{Base-A} \qquad \frac{x \in X(\sigma) \qquad Q(x) = a}{Reach(\sigma, i, a)}\ \text{Base-B} \qquad \frac{\sigma \vdash a \xrightarrow{f} b \qquad Reach(\sigma, i, a)}{Reach(\sigma, i, b)}\ \text{Step}$$

Theorem B.7 (Name). *Statement.*

Theorem B.8 (Name). *Let σ be such that $P(\sigma)$. Then*

$$Q(\sigma) \implies R(\sigma).$$

References

Hoang Nguyen. 2026. *Verona through Rome - Revisiting Verona's model and region system.* Technical Report. Imperial College London. https://github.com/hxcuber/verona-through-rome/blob/main/report.pdf